Radar HRRP Target Recognition Based on Higher Order Spectra

Lan Du, Hongwei Liu, *Member, IEEE*, Zheng Bao, *Senior Member, IEEE*, and Mengdao Xing

***Abstract*—Radar high-resolution range profile (HRRP) is very sensitive to time-shift and target-aspect variation; therefore, HRRP-based radar automatic target recognition (RATR) requires efficient time-shift invariant features and robust feature templates. Although higher order spectra are a set of well-known time-shift invariant features, direct use of them (except for power spectrum) is impractical due to their complexity. A method for calculating the Euclidean distance in higher order spectra feature space is proposed in this paper, which avoids calculating the higher order spectra, effectively reducing the computation complexity and storage requirement. Moreover, according to the widely used scattering center model, theoretical analysis and experimental results in this paper show that the feature vector extracted from the average profile in a small target-aspect sector has better generalization performance than the average feature vector in the same sector when both of them are used as feature templates in HRRP-based RATR. The proposed Euclidean distance calculation method and average profile-based template database are applied to two classification algorithms [the template matching method (TMM) and the radial basis function network (RBFN)] to evaluate the recognition performances of higher order spectra features. Experimental results for measured data show that the power spectrum has the best recognition performance among higher order spectra.**

***Index Terms*—High-resolution range profile (HRRP), higher order spectra, radar automatic target recognition (RATR), radial basis function network (RBFN), template matching method (TMM), time-shift invariant feature.**

I. Introduction

A high-resolution range profile (HRRP) is the amplitude of the coherent summations of the complex time returns from target scatterers in each range resolution cell, which represents the projection of the complex returned echoes from the target scattering centers onto the radar line of sight (LOS). It contains the target structure signatures, such as target size, scatterer distribution, etc. Therefore, radar HRRP target recognition has received intensive attention from the radar automatic target recognition (RATR) community [1]–[12]. Several issues must be considered when HRRP is applied to radar target recognition. The first one is the well-known target-aspect sensitivity of HRRP. The scattering center model is widely used by the high-range resolution radar imaging community, which has been proven to be a simple and efficient target model in synthetic aperture radar (SAR) imaging and inverse synthetic aperture radar (ISAR) imaging [13]. According to this model, the variation of the target aspect will lead to different range shifts for different scattering centers on the target, even within the aspect region where the scattering center structure remains unchanged; therefore, the HRRP, which is the amplitude of coherent sum of the complex returned echoes from scatterers in a range resolution cell, can be changed substantially. The second one is the time-shift sensitivity of HRRP. In order to decrease the computation complexity, HRRP is only a part of received radar echo extracted by a range window, in which a target signal is included. Thus, the position of the target signal in HRRP can vary with the measurement. The last one is the amplitude sensitivity of HRRP. This comes from the fact that the amplitude of HRRP is a function of target distance, radar antenna gain, radar receiver gain, and so on. Therefore, directly matching a test HRRP with a template is a three-dimensional (3-D) search problem with a huge computation burden. How to reduce the computation complexity and achieve good generalization performance is a challenging problem in HRRP-based RATR.

As a time-shift invariant feature, bispectra have been well studied by the RATR community [8], [11], [12], [14], [15]. The direct use of the bispectra feature results in a two-dimenional (2-D) matching score that is not available for real RATR systems due to their huge computation burden. To overcome this difficulty, some dimensionality-reduction methods [8], [11], [12], [14], [15] were proposed. Actually, higher order spectra are a set of time-shift invariant features, among which, bispectra, i.e., the second-order spectra, is just a particular case of higher order spectra. The reason why the other nth-order spectra features ($n > 2$) receive less attention from RATR community is that the storage requirement is too huge, and the computation is too complex. In this paper, a method for calculating the Euclidean distances in higher order spectra feature space is proposed. This method is performed in original HRRP space directly, which avoids calculating the higher order spectra; therefore, the computation complexity and storage requirement can be extremely decreased.

Due to the HRRP's target-aspect sensitivity mentioned above, one simple approach to build an efficient RATR system is to establish a template database, including HRRPs from all target aspects. However, this approach is impractical because of its huge computation burden and storage requirement. Several literatures [6], [8]–[10] showed that the average profile in a small target-aspect sector can represent all the profiles in the sector well. There-

Manuscript received January 12, 2004; revised July 23, 2004. This work was supported in part by the National Science Foundation of China under Grant 60302009 and the National Defense Preresearch Foundation of China under Grant 413070501. The associate editor coordinating the review of this manuscript and approving it for publication was Prof. Alex B. Gershman.

L. Du, Z. Bao, and M. Xing are with the National Lab of Radar Signal Processing, Xidian University, Xi'an, China (e-mail: dulan@mail.xidian.edu.cn).

H. Liu was with the Department of Electrical and Computer Engineering, Duke University, Durham, NC 27708 USA. He is now with the National Lab of Radar Signal Processing, Xidian University, Xi'an, China.

Digital Object Identifier 10.1109/TSP.2005.849161

1053-587X/$20.00 © 2005 IEEE

fore, when RATR is performed in HRRP space, average profiles in all target-aspect sectors can be used as templates. However, for the case of performing RATR in a time-shift invariant feature space, how to establish robust feature templates remains unclear. According to the scattering center model, theoretical analysis and experimental results in this paper show that the feature vector extracted from the average profile in a small target-aspect sector has better generalization performance than the average feature vector in the same sector when both of them are used as feature templates. Thus, in order to achieve good generalization performance, it is better to use feature vectors extracted from average profiles as feature templates rather than using average feature vectors.

The proposed Euclidean distance calculation method and average profile-based template database are applied to two classification algorithms [template matching method (TMM) and radial basis function network (RBFN)] to evaluate the recognition performances of higher order spectra features. Experimental results for measured data show that the power spectrum has the best recognition performance among higher order spectra, including well-studied bispectra.

This paper is organized as follows. In Section II, a computationally efficient method is proposed to calculate the Euclidean distances in higher order spectra feature space, with some detailed formula derivation deferred to the Appendix . In Section III, the problem of how to establish the template database with good generalization performance is discussed. In Section IV, TMM and RBFN are introduced. In Section V, recognition experiments based on measured data are performed to evaluate the recognition performances of higher order spectra features. Finally, conclusions are made in Section VI.

II. Euclidean Distances in Higher Order Spectra Feature Space

A. Definition of Higher Order Spectra

Let $x(t)$ be a real and continuous function of time t. The nth-order spectra of $x(t)$ is defined as

$$\begin{aligned} &F_{xx}^{(n)}(\omega_1,\ldots,\omega_n) \\ &\triangleq \int\cdots\int r_{xx}^{(n)}(\tau_1,\ldots,\tau_n)e^{-j(\omega_1\tau_1+\cdots+\omega_n\tau_n)}d\tau_1\cdots d\tau_n \\ &= \int\cdots\int\left[\int x(t)\prod_{k=1}^{n}x(t+\tau_k)dt\right]\cdot e^{-j(\omega_1\tau_1+\cdots+\omega_n\tau_n)}d\tau_1\cdots d\tau_n \end{aligned} \tag{1}$$

where $r_{xx}^{(n)}(\tau_1,\ldots,\tau_n)$ is the nth-order autocorrelation function of $x(t)$.[1] Equation (1) can also be rewritten as

$$F_{xx}^{(n)}(\omega_1,\ldots,\omega_n) = X(\omega_1)\cdots X(\omega_n)X^*\left(\sum_{k=1}^{n}\omega_k\right). \tag{2}$$

[1]In much of the literature, higher order spectra are defined as the Fourier transformation of higher order cumulants [16], [17], which is different from the higher order spectra defined in this paper due to the nonzero mean of HRRP. However, higher order spectra defined in this paper are more often used in HRRP-based RATR [8], [11], [12]. A fact worth paying attention to is that this kind of higher order spectra does not always have the characteristics of higher order cumulants.

where $*$ denotes the complex conjugate. Especially, $F_{xx}^{(1)}(\omega)$ is the power spectrum, $F_{xx}^{(2)}(\omega_1,\omega_2)$ is the bispectra, and $F_{xx}^{(3)}(\omega_1,\omega_2,\omega_3)$ is the trispectra of $x(t)$. Let $x_\tau(t) = x(t-\tau)$. It can be easily proved that $F_{x(t)x(t)}^{(n)}(\omega_1,\ldots,\omega_n) = F_{x_\tau(t)x_\tau(t)}^{(n)}(\omega_1,\ldots,\omega_n)$. Obviously, the higher order spectra are independent of the HRRP's time-shift, which is referred to as the time-shift invariant property.

B. Euclidean Distance in Higher Order Spectra Feature Space

The Euclidean distance is widely used in pattern recognition to measure the similarity between patterns. According to template matching method under the maximum correlation coefficient criterion (MCC-TMM) [2], [3], [8], [10] (there is a detailed illustration shown in Section IV A), given two HRRPs $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$, their squared Euclidean distance is calculated by searching the optimal time-shift τ such that the minimum distance is reached, i.e.,

$$\begin{aligned} d_{\mathrm{E}}^2(\overline{\boldsymbol{x}}_1,\overline{\boldsymbol{x}}_2) &= \min_{\tau}\|\overline{\boldsymbol{x}}_1-\overline{\boldsymbol{x}}_2(\tau)\|_2^2 = 2(1-\max_{\tau}\langle\overline{\boldsymbol{x}}_1,\overline{\boldsymbol{x}}_2(\tau)\rangle) \\ &= 2(1-\max_{\tau}\overline{r}_{x_1x_2}^{(1)}(\tau)) \end{aligned} \tag{3}$$

where $\overline{\boldsymbol{x}}_1$ and $\overline{\boldsymbol{x}}_2$, respectively, represent the L_2 normalized vectors [2], [3], [8], [10] of $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$, and $\overline{r}_{x_1x_2}^{(1)}$ is the amplitude normalized cross-correlation function between $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$. Define the time-shift compensated cross-correlation coefficient between $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ as

$$\rho_{x_1x_2} \triangleq \max_{\tau}\overline{r}_{x_1x_2}^{(1)}(\tau) = \max_{\tau}\frac{r_{x_1x_2}^{(1)}(\tau)}{\sqrt{r_{x_1x_1}^{(1)}(0)\cdot r_{x_2x_2}^{(1)}(0)}}. \tag{4}$$

Substituting (4) into (3), we get

$$d_{\mathrm{E}}^2(\overline{\boldsymbol{x}}_1,\overline{\boldsymbol{x}}_2) = 2(1-\rho_{x_1x_2}). \tag{5}$$

Obviously, since the Euclidean distance is a monotonous function of the cross-correlation coefficient, the Euclidean distance and the cross-correlation coefficient are equivalent when used as similarity measures between two patterns.

Now, we consider calculation of the Euclidean distance between two feature vectors in higher order spectra feature space. Since higher order autocorrelation function and higher order spectra are time-shift invariant, time-shift compensation need not be considered. It is easy to prove that Euclidean distance is a monotonous function of the inner product. By formula derivation [18] (as shown in the Appendix), the inner product of two autocorrelation functions is

$$\begin{aligned} \left\langle \boldsymbol{r}_{x_1x_1}^{(n)},\boldsymbol{r}_{x_2x_2}^{(n)}\right\rangle &= \int\left[\int x_1(t)x_2(t+\tau)dt\right]^{n+1}d\tau \\ &= \int\left[r_{x_1x_2}^{(1)}(\tau)\right]^{n+1}d\tau \end{aligned} \tag{6}$$

where $\boldsymbol{r}_{x_1x_1}^{(n)}$ and $\boldsymbol{r}_{x_2x_2}^{(n)}$ represent the nth-order autocorrelation functions of $x_1(t)$ and $x_2(t)$, respectively. In (6), the inner product of $\boldsymbol{r}_{x_1x_1}^{(n)}$ and $\boldsymbol{r}_{x_2x_2}^{(n)}$ is the integral of $\left[r_{x_1x_2}^{(1)}(\tau)\right]^{n+1}$ with respect to time-shift τ; thus, it is independent of the time-shift τ. Considering the amplitude normalization, similar to (4), the cross-correlation coefficient between $\boldsymbol{r}_{x_1x_1}^{(n)}$ and $\boldsymbol{r}_{x_2x_2}^{(n)}$ is defined as

$$\begin{aligned}\rho_{\boldsymbol{r}_{x_1x_1}^{(n)}\boldsymbol{r}_{x_2x_2}^{(n)}} &\triangleq \left\langle \overline{\boldsymbol{r}}_{x_1x_1}^{(n)}, \overline{\boldsymbol{r}}_{x_2x_2}^{(n)} \right\rangle = \frac{\left\langle \boldsymbol{r}_{x_1x_1}^{(n)}, \boldsymbol{r}_{x_2x_2}^{(n)} \right\rangle}{\sqrt{\left\langle \boldsymbol{r}_{x_1x_1}^{(n)}, \boldsymbol{r}_{x_1x_1}^{(n)} \right\rangle \cdot \left\langle \boldsymbol{r}_{x_2x_2}^{(n)}, \boldsymbol{r}_{x_2x_2}^{(n)} \right\rangle}} \\ &= \frac{\int \left[r_{x_1x_2}^{(1)}(\tau) \right]^{n+1} d\tau}{\sqrt{\int \left[r_{x_1x_1}^{(1)}(\tau) \right]^{n+1} d\tau \cdot \int \left[r_{x_2x_2}^{(1)}(\tau) \right]^{n+1} d\tau}} \\ &= \int \tilde{\gamma}_{x_1x_2}^{(1)}|_{(n+1)}(\tau) d\tau \end{aligned} \tag{7}$$

where

$$\tilde{\gamma}_{x_1x_2}^{(1)}|_{(n+1)}(\tau) = \left[\frac{r_{x_1x_2}^{(1)}(\tau)}{\left(\sqrt{\int \left[r_{x_1x_1}^{(1)}(\tau) \right]^{n+1} d\tau \cdot \int \left[r_{x_2x_2}^{(1)}(\tau) \right]^{n+1} d\tau} \right)^{1/(n+1)}} \right]^{n+1}$$

is called the nth-order area normalized cross-correlation function between $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$. Because higher order autocorrelation functions can be transformed into higher order spectra by the Fourier transformation, which is an orthogonal transformation, the relation between their inner products is given by

$$\left\langle \boldsymbol{F}_{x_1x_1}^{(n)}, \boldsymbol{F}_{x_2x_2}^{(n)} \right\rangle = (2\pi)^n \left\langle \boldsymbol{r}_{x_1x_1}^{(n)}, \boldsymbol{r}_{x_2x_2}^{(n)} \right\rangle. \tag{8}$$

Then, the cross-correlation coefficient between the two higher order spectra is

$$\begin{aligned}\rho_{\boldsymbol{F}_{x_1x_1}^{(n)}\boldsymbol{F}_{x_2x_2}^{(n)}} &= \frac{\left\langle \boldsymbol{F}_{x_1x_1}^{(n)}, \boldsymbol{F}_{x_2x_2}^{(n)} \right\rangle}{\sqrt{\left\langle \boldsymbol{F}_{x_1x_1}^{(n)}, \boldsymbol{F}_{x_1x_1}^{(n)} \right\rangle, \left\langle \boldsymbol{F}_{x_2x_2}^{(n)}, \boldsymbol{F}_{x_2x_2}^{(n)} \right\rangle}} \\ &= \frac{\left\langle \boldsymbol{r}_{x_1x_1}^{(n)}, \boldsymbol{r}_{x_2x_2}^{(n)} \right\rangle}{\sqrt{\left\langle \boldsymbol{r}_{x_1x_1}^{(n)}, \boldsymbol{r}_{x_1x_1}^{(n)} \right\rangle \cdot \left\langle \boldsymbol{r}_{x_2x_2}^{(n)}, \boldsymbol{r}_{x_2x_2}^{(n)} \right\rangle}} = \rho_{\boldsymbol{r}_{x_1x_1}^{(n)}\boldsymbol{r}_{x_2x_2}^{(n)}}. \end{aligned} \tag{9}$$

Therefore

$$d_{\mathrm{E}}^2\left(\overline{\boldsymbol{F}}_{x_1x_1}^{(n)}, \overline{\boldsymbol{F}}_{x_2x_2}^{(n)}\right) = d_{\mathrm{E}}^2\left(\overline{\boldsymbol{r}}_{x_1x_1}^{(n)}, \overline{\boldsymbol{r}}_{x_2x_2}^{(n)}\right) = 2\left(1 - \rho_{\boldsymbol{r}_{x_1x_1}^{(n)}\boldsymbol{r}_{x_2x_2}^{(n)}}\right). \tag{10}$$

Similarly, the Euclidean distance in higher order spectra feature space and the cross-correlation coefficient in the higher order autocorrelation function space are equivalent when used as similarity measures between two patterns.

The nth-order spectra feature of an HRRP with a length of M is with the dimension of $\underbrace{M \times M \cdots \times M}_{n}$. When either the length of the HRRP or the order of the nth-order spectra feature increases, the computation complexity and storage requirement of the nth-order spectra feature will increase rapidly. Therefore, the nth-order spectra features ($n \geq 2$) are not practical for real RATR systems. Some dimensionality reduction methods were proposed for the bispectra feature [8], [11], [12], [14], [15]. When each of these methods is used, bispectra features need to be calculated in the preprocessing step, which is still a large computation burden. To our knowledge, there is no dimensionality reduction method for the nth-order spectra features ($n > 2$) due to their complexity. However, by (10), the Euclidean distance in higher order spectra feature space can be calculated in the original HRRP space directly, which avoids calculating higher order spectra features; therefore, the computation complexity and storage requirement can be extremely decreased.

III. Establishment of Template Database

As mentioned above, since HRRP has the target-aspect sensitivity, it is impractical to store HRRPs from all target-aspects as templates. Therefore, how to establish a robust template database is a very important issue in HRRP-based RATR, which will be discussed in this section.

A. Importance of Generalization Performance

Given a set of training data, its clustering center is generally taken as the template. If the Euclidean distance is used to measure the similarity between data samples, the clustering center can be obtained by the minimum mean square error (MSE) criterion. Let $\{\overline{\boldsymbol{X}}_1, \overline{\boldsymbol{X}}_2, \ldots, \overline{\boldsymbol{X}}_K\}$ be a set of amplitude normalized (L_2 normalization [2], [3], [8], [10]) training data, and let $\boldsymbol{X}$ be the clustering center to be calculated. For the case that the prior probabilities of the training data are assumed to be equal, the optimization problem becomes

$$\min_{\boldsymbol{X}} d_{\mathrm{E}}^2 = \min_{\boldsymbol{X}} \sum_{i=1}^{K} \left\| \overline{\boldsymbol{X}}_i - \boldsymbol{X} \right\|_2^2. \tag{11}$$

The least square solution of (11) is

$$\boldsymbol{X} = \frac{1}{K} \sum_{i=1}^{K} \overline{\boldsymbol{X}}_i. \tag{12}$$

Therefore, mathematically speaking, the average feature vector is the optimal solution of the clustering center under the Euclidean distance criterion.

Using average feature vectors as templates can minimize the empirical risk but cannot guarantee good generalization performance, especially for a small training data set. For HRRP-based RATR, the training data set available is incomplete. In this case, generalization performance is one of the most important factors worth paying attention to. Recently, the support vector machine (SVM) [19], which is an efficient machine learning algorithm aimed at minimizing structural risk for good generalization performance, has received intensive attention from the pattern recognition community. Nevertheless, a fact worth pointing out is that the SVM is a data-driven machine learning algorithm in which only mathematical constraint is used in the training stage, regardless of any target's physical property. For HRRP-based RATR, we will show that the target's physical structure information can be used to establish a robust feature template database.

B. Physical Mechanism of Average Profiles

Generally, since targets and their components are much larger than the wavelength of microwave radar, the electromagnetism characteristics of the targets can be described by the scattering center target model [13]. As mentioned above, compared with the rapid change of the scatterer distribution, the scattering center model varies much more slowly with the target-aspect. As shown in Fig. 1, the target rotates around the point O at a uniform angular speed vector $\boldsymbol{\Omega}$, and $\boldsymbol{R}$ represents the unit vector from the target to the radar. It is easy to see if the angular speed vector $\boldsymbol{\Omega}$ is decomposed into a vector $\boldsymbol{\Omega}_R$ paralleling to $\boldsymbol{R}$ and a vector $\boldsymbol{\Omega}_e$ perpendicular to $\boldsymbol{R}$, only the vector $\boldsymbol{\Omega}_e$, which is referred to as the effective rotation vector, can cause the Doppler effect. Let A be a target scatterer, and let $\boldsymbol{r}_A$ represent its position vector from the point O. Thus, the instantaneous Doppler frequency of the scatterer A is

$$f_d = \frac{2}{\lambda}(\boldsymbol{\Omega}_e \times \boldsymbol{r}_A \bullet \boldsymbol{R}) = \frac{2}{\lambda}(\boldsymbol{\Omega}_e \times \boldsymbol{R} \bullet \boldsymbol{r}_A) \tag{13}$$

where λ is the wavelength, the vector ($\boldsymbol{\Omega}_e \times \boldsymbol{R}$) is perpendicular to the $\boldsymbol{\Omega}_e - \boldsymbol{R}$ plane, and f_d is proportional to the projection of $\boldsymbol{r}_A$ onto the vector ($\boldsymbol{\Omega}_e \times \boldsymbol{R}$). Therefore, the 3-D rotation of target can be projected onto the 2-D plane spanned by $\boldsymbol{R}$ and ($\boldsymbol{\Omega}_e \times \boldsymbol{R}$), which, respectively, correspond to radial range and cross range dimension in an ISAR image. The limitation of the target-aspect change to avoid scatterers' motion through resolution cells (MTRC) is given by [8]–[10], [13]

$$\Delta\varphi = \Omega_e \cdot \Delta t \leq \frac{\Delta R}{L} \tag{14}$$

where ΔR is the length of range resolution cell, and L is the cross length of the target. Since the scattering center model almost remains unchanged within $\Delta\varphi$, the HRRPs in this sector can be regarded as a vector stationary process. HRRPs of one target can be divided into many subsets according to different angular sectors, which are defined as HRRP frames in this paper.

If a target uniformly rotates within the target-aspect sector corresponding to an HRRP frame, the mth complex returned echo ($m = 0, 1, \ldots, M-1$) in the nth range cell ($n = 1, 2, \ldots, N$) is

$$x_n(m) = \sum_{i=1}^{L_n} \sigma_{ni} e^{-j[(4\pi/\lambda)\Delta r_i(m) - \psi_{i0}]} = \sum_{i=1}^{L_n} \sigma_{ni} e^{j\phi_{ni}(m)} \tag{15}$$

where λ is the wavelength, L_n denotes the number of target scatterers in the nth range cell, $\Delta r_i(m)$ represents the radial displacement of the ith scatterer in the mth returned echo, and σ_{ni} and ψ_{i0} represent the strength and the initial phase of the ith scatterer, respectively. Then, the HRRP of the mth time return is defined as

$$\boldsymbol{x}(m) \triangleq [|x_1(m)|, |x_2(m)|, \ldots, |x_N(m)|]^T. \tag{16}$$

The power of $x_n(m)$ can be written as

$$\begin{aligned} |x_n(m)|^2 &= x_n(m) \cdot x_n^*(m) \\ &= \sum_{i=1}^{L_n} \sigma_{ni}^2 + 2\sum_{i=2}^{L_n}\sum_{k=1}^{i-1} \sigma_i \sigma_k \cos[\theta_{nik}(m)] \end{aligned} \tag{17}$$

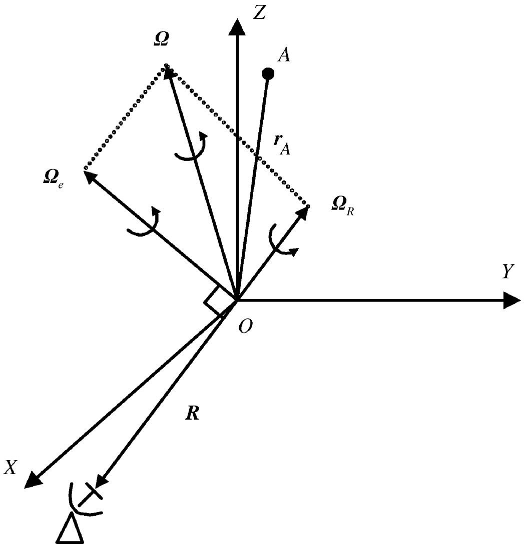

Fig. 1. Target's 3-D rotation sketch map: The target rotates around the point O at a uniform angular speed vector $\boldsymbol{\Omega}$, and $\boldsymbol{R}$ represents the unit vector from the target to the radar. The vectors $\boldsymbol{\Omega}_R$ and $\boldsymbol{\Omega}_e$ are two components of $\boldsymbol{\Omega}$, of which $\boldsymbol{\Omega}_R$ is parallel to $\boldsymbol{R}$, and $\boldsymbol{\Omega}_e$ is perpendicular to $\boldsymbol{R}$. A represents a target scatterer, and $\boldsymbol{r}_A$ is its position vector from the point O.

where $\theta_{nik}(m)$ represents the phase difference between the ith scatterer and the kth scatterer of the mth echo in the nth range cell.

$$\begin{aligned} \theta_{nik}(m) &= \phi_{ni}(m) - \phi_{nk}(m) \\ &= (\psi_{i0} - \psi_{k0}) - \frac{4\pi}{\lambda}[\Delta r_i(m) - \Delta r_k(m)] \\ &= \theta_{nik}(0) + \delta\theta_{nik}(m). \end{aligned} \tag{18}$$

The first term at the right side of (17) is referred to as the scatterer auto-term (SAT), which is the summation of the scatterers' energy within a range cell; the second term is referred to as the scatterer cross-term (SCT), which is the summation of the conjugate multiplication of echoes from different scatterers within a range cell. Without MTRC, the SAT remains invariant, but the SCT varies as a random variable with zero mean [8], [10]. Thus, the target-aspect sensitivity of HRRP results from the SCT. In order to reduce the target-aspect sensitivity for HRRP, the objective is in fact the reduction of SCT. The average profile of the HRRP frame is defined as

$$\hat{\boldsymbol{x}} \triangleq \left[\sqrt{\frac{1}{M}\sum_{m=0}^{M-1}|x_1(m)|^2}, \sqrt{\frac{1}{M}\sum_{m=0}^{M-1}|x_2(m)|^2}, \ldots, \sqrt{\frac{1}{M}\sum_{m=0}^{M-1}|x_N(m)|^2}\right]^T. \tag{19}$$

Due to the time-shift sensitivity of HRRP, to obtain the average profile, the range alignment should be performed to the set of HRRPs first; in other words, the average profile $\hat{\boldsymbol{x}}$ is obtained from range aligned HRRPs $\{\boldsymbol{x}(0), \boldsymbol{x}(1), \ldots, \boldsymbol{x}(M-1)\}$ [8], [10]. We now show that the average profile can reduce the SCT in the HRRP frame to a large degree.

Let $\tilde{\boldsymbol{x}}(m) = [x_1(m), x_2(m), \ldots, x_N(m)]^T$ represent the complex HRRP of the mth time return. Performing the Fourier transformation to each row of $\tilde{\boldsymbol{x}} = [\tilde{\boldsymbol{x}}(0), \tilde{\boldsymbol{x}}(1), \ldots, \tilde{\boldsymbol{x}}(M-1)]$,

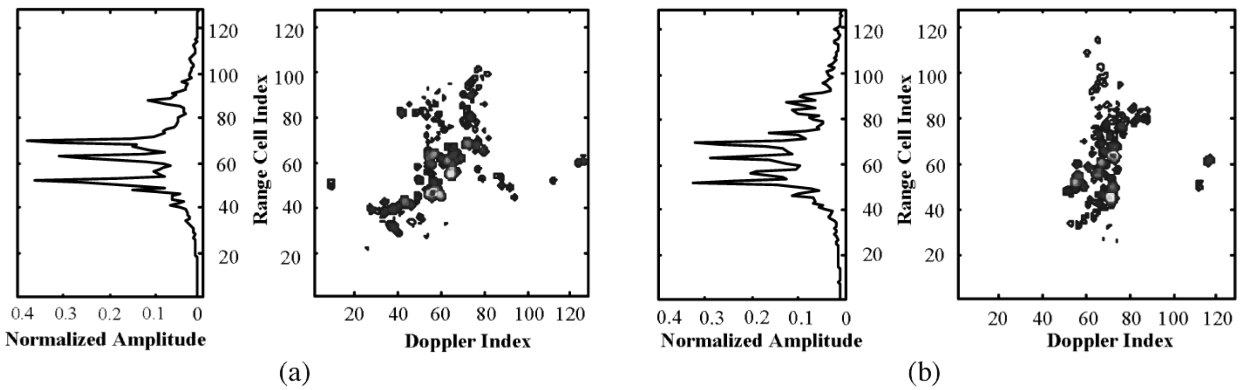

Fig. 2. Instantaneous ISAR images and their average profiles of "Yark-42" in the maneuvering flight. (a) $t = 0.3$ s ($0 \sim 0.6$ s). (b) $t = 4.75$ s ($4.25 \sim 5.25$ s).

which corresponds to each range cell, one can obtain the ISAR image $\boldsymbol{X} = [\boldsymbol{X}(0), \boldsymbol{X}(1), \ldots, \boldsymbol{X}(M-1)]$, where $\boldsymbol{X}(k) = [X_1(k), X_2(k), \ldots, X_N(k)]^T$, and $X_n(k)$ represents the complex strength summation of the scatterers in the resolution gridding (n, k), which is referred to as an image pixel. Obviously, an ISAR image is the 2-D distribution of the target's scatterer energy, where scatterers are resolved to some extent in the cross range dimension as well as in the radial range dimension. When a target rotates within the target-aspect sector corresponding to an HRRP frame, the direction of the effective rotation vector $\boldsymbol{\Omega}_e$ may display drastic changes, and thus, the cross-range distribution of scatterers in each range cell changes greatly, as is seen from (13). On the other hand, the amount of change in orientation (including aspect and elevation) of the target is usually small during this short period, and thus, the distribution of scatterers in the radial range dimension (and, consequently, the projection of the ISAR image onto the radial range dimension) will remain quite stable. The stable radial-range projection of the ISAR image is identical to the summation of the power profiles of the HRRP frame according to the energy preservation property of the Fourier transformation, i.e., the Paseval theorem

$$\sum_{k=0}^{M-1} |X_n(k)|^2 = \sum_{m=0}^{M-1} |x_n(m)|^2. \tag{20}$$

Moreover, the term at the left side of (20), which is the energy summation of the image pixels within the nth range cell, is also equal to the stable SAT in the nth range cell approximately; the one at the right side (the summation of all range aligned power echoes in the nth range cell) is identical to the average power profile after being amplitude normalized.[2] Therefore, the average profile defined in (19) can represent the stable SAT approximately and reduce the SCT in the HRRP frame to a large degree. To make this point clearer, we give two ISAR images of a maneuvering airplane along with their average profiles in Fig. 2 [10], [20]. The ISAR image in Fig. 2(a) is obtained by using the data collected in the time interval $0 \sim 0.6$ s and slicing the time-frequency distribution at the time instant of 0.3 *s*, whereas the ISAR image in Fig. 2(b) is obtained by using the data collected in the time interval $4.25 \sim 5.25$ s and slicing the time-frequency distribution at the time instant of 4.75 s. Although the time interval between the two images is no more than 5 s, the two ISAR images have significant differences in cross-range distribution of scatterers due to the target's instantaneous maneuver property. However, the scatterer distribution in the radial-range dimension, i.e., the SAT, basically remains unchanged. Therefore, the two average profiles shown in Fig. 2, which are, respectively, obtained by averaging power profiles over the time interval of $0 \sim 0.6$ s and $4.25 \sim 5.25$ s, are very similar, especially where the positions of peaks are almost fixed. This demonstrates that the average profile can describe the scattering center model within an aspect sector well.

C. Comparison Between Feature Vectors Extracted From Average Profiles and Average Feature Vectors

As mentioned above, since the training data set available in radar HRRP target recognition is incomplete, generalization performance is one of the most important factors worth paying attention to. Due to the sound physical mechanism of average profiles, feature vectors extracted from the average profiles can represent the target's stable physical structure information, but average feature vectors do not have this property. Therefore, although the average feature vectors are the mathematically optimal templates, as far as generalization performance is concerned, feature vectors extracted from the average profiles are more suitable to be feature templates.

In order to compare the generalization performances of the two kinds of feature templates, two sets of data with different elevation angles are simulated by a radar backscattering simulation software [21], [22]. The data with about a 0° elevation angle are used as training data to establish templates, and the other data with about 3° elevation angle are used as test data. The parameters of targets and radar are shown in Table I. The target-aspect change between the two neighboring HRRPs is about 0.01°. According to (14) and the radar parameters shown in Table I, and in view of a few scatterers distributed at the edge of an aircraft-like target and with weak energy, HRRPs

[2]In the above analysis, we assume that the complex HRRPs are range aligned and phase compensated. Thus, using the Fourier transformation can obtain a focused ISAR image; in other words, the multiscatterers in every range cell can be resolved approximately [10], [20]. However, because the term at the left side of (20) is ***energy*** summation, phase adjustment will not affect it; in other words, the term at the left side remains invariant whether the complex HRRPs are phase compensated or not. Thereby, as stated following (19), to obtain an average profile only needs to perform range alignment first.

TABLE I
PARAMETERS OF PLANES AND RADAR IN THE SIMULATED EXPERIMENT

radar parameters	center frequency		5520 MHz
	bandwidth		400 MHz
planes	length (m)	width (m)	scale
Tu-16	34.80	33.00	2:1
B-52	49.50	56.40	2:1
B-1B	44.80	23.80	2:1
Tornado	16.72	13.91	1:1
Mig-21	15.76	7.15	1:1
F-15	19.43	13.05	1:1

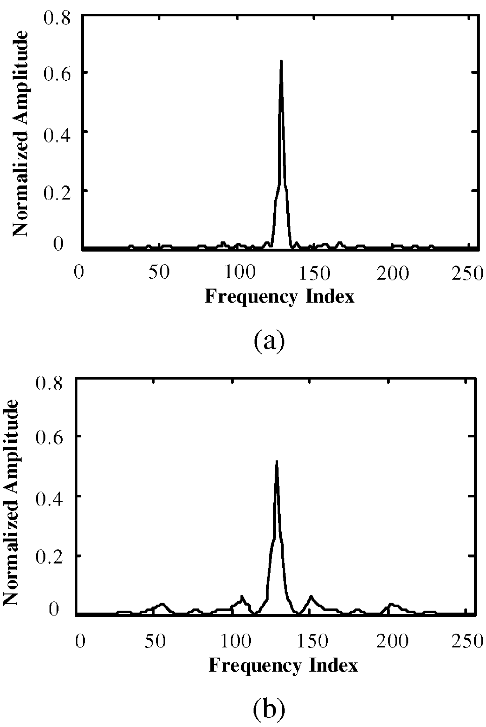

Fig. 3. Comparison between the waveforms of a power spectrum extracted from an average profile and an average power spectrum. (a) Power spectrum extracted from an average profile. (b) Average power spectrum.

TABLE II
COMPARISON IN RECOGNITION PERFORMANCE BETWEEN THE TWO KINDS OF FEATURE TEMPLATES

recognition rates by MCC-TMM (%) / the azimuth angles of test data	power spectrum vectors extracted from the average profiles as feature templates	average power spectrum vectors as feature templates	the recognition rate differences
S1:30~20°	89.08	85.58	3.50
S2:40~30°	89.43	87.80	1.63
S3:50~40°	90.58	87.90	2.69
S4:60~50°	80.12	79.25	0.87
S5:70~60°	67.68	66.97	0.71
S6:120~110°	87.72	83.67	4.05
S7:130~120°	91.85	89.63	2.22
S8:140~130°	96.13	94.45	1.69
S9:150~140°	91.38	88.98	2.40
S10:160~150°	91.57	87.08	4.49
S1~S10	87.55	85.13	2.42

in about a 3° angular sector are taken as an HRRP frame in our experiments. Fig. 3 shows a comparison between the waveforms of a power spectrum vector extracted from an average profile and an average power spectrum vector, which both are obtained from one HRRP frame. Because the average profile can suppress the amplitude fluctuation property caused by the SCT, the spectrum strengths corresponding to the higher frequency components of its power spectrum are less than those of the average power spectrum. Table II shows a comparison of the recognition performance between the two kinds of feature templates by using MCC-TMM as a classification algorithm (regarding MCC-TMM, a detailed illustration is shown in Section IV-A). Obviously, the recognition rates obtained by using power spectrum vectors extracted from average profiles as feature templates are larger than those obtained by using average power spectrum vectors as feature templates, especially for some test segments where the differences are larger than four percentage points. For the measured data (for a detailed illustration, see Section V-A), the average recognition rate difference by using MCC-TMM as a classification algorithm is about two percentage points. Although the differences in recognition performance between the two template database establishment methods appear to be minor, they are statistically significant according to the statistical tests proposed in [23]. The conclusion drawn from the power spectrum can be extended to higher order spectra.

However, one point worth emphasizing is that we cannot say that using the average feature vectors as feature templates are inferior in any case. Theoretically speaking, if the training data set were complete, the recognition performance of using average feature vectors as feature templates would be better than that of using feature vectors extracted from average profiles as feature templates. It is the incompleteness of the training data set available in the radar HRRP target recognition that leads to the superiority of feature vectors extracted from average profiles as feature templates. The conclusion that we can draw from the above analysis and experiments is that using the target's physical structure information to establish templates is suitable for the incomplete training data set to achieve good generalization performance, which is especially important for radar HRRP target recognition due to the target-aspect sensitivity of HRRP.

IV. CLASSIFICATION ALGORITHMS

Classification algorithms also play an important role in radar HRRP target recognition. Two classification algorithms (TMM and RBFN) are applied to evaluate the recognition performances of higher order spectra features.

A. Template Matching Method (TMM)

A fundamental method in pattern recognition is TMM [24]–[26], in which the class of a test data is determined according to matching scores between it and all templates.

The TMM under the maximum correlation coefficient (MCC) criterion is referred to as the MCC-TMM.

Let $\boldsymbol{x}$ be a test data. The nth-order spectra cross-correlation coefficients between $\boldsymbol{x}$ and all the average profile templates $\{\overline{\boldsymbol{x}}_{i,k}, k = 1, 2, \ldots, K_i\}$ of target $T_i (i = 1, 2 \ldots, M)$ are calculated. First, the MCC of each target is searched along the aspect axis.

$$\rho_{\boldsymbol{F}^{(n)}_{xx} \boldsymbol{F}^{(n)}_{\overline{x}_i \overline{x}_i}} = \max_k \rho_{\boldsymbol{F}^{(n)}_{xx} \boldsymbol{F}^{(n)}_{\overline{x}_{i,k} \overline{x}_{i,k}}}, \quad k = 1, 2, \ldots, K_i \tag{21}$$

where $\rho_{\boldsymbol{F}^{(n)}_{xx} \boldsymbol{F}^{(n)}_{\overline{x}_i \overline{x}_i}}$ is taken as the nth-order spectra cross-correlation coefficient between $\boldsymbol{x}$ and target T_i. Then, the class of the test data can be determined by searching the maximum value among the MCCs of all targets. If

$$j = \arg\max_i \rho_{\boldsymbol{F}^{(n)}_{xx} \boldsymbol{F}^{(n)}_{\overline{x}_i \overline{x}_i}}, \quad i = 1, 2 \ldots, M \tag{22}$$

$\boldsymbol{x}$ belongs to target T_j.

B. Radial Basis Function Network (RBFN)

RBFN is a well-known kernel function classifier [8], [12], [24]–[26]. A Gaussian RBFN with K centers has the form

$$g(\boldsymbol{x}) = \sum_{k=1}^{K} w_k \mathbb{F}_k(\boldsymbol{x}) + b = \sum_{k=1}^{K} w_k \exp\left(-\frac{\|\boldsymbol{x} - \boldsymbol{c}_k\|^2}{2\sigma_k^2}\right) + b \tag{23}$$

where $\mathbb{F}_k$ is the kth Gaussian basis function with center $\boldsymbol{c}_k$ and variance σ_k^2. The weight coefficients $w_k (k = 1, 2, \ldots, K)$ combine the Gaussian terms into a single output value, and b is a bias term. Building a Gaussian RBFN for a given learning task needs to determine the parameters, including the basis function centers, variances, weights, bias term, and center number. The detailed training and test procedure of RBFN can be found in [12], [26]. The main differences between the method used in this paper and the traditional method are summarized as follows.

In this paper, the number of Gaussian basis functions equals the number of HRRP frames. The center $\boldsymbol{c}_k^{(n)}$ corresponding to the nth-order spectra features of the kth HRRP frame is

$$\boldsymbol{c}_k^{(n)} = \boldsymbol{F}^{(n)}_{\overline{x}_k \overline{x}_k} \tag{24}$$

and its variance $\sigma_k^2|^{(n)}$ is

$$\sigma_k^2|^{(n)} = \frac{1}{L} \sum_{l=1}^{L} \left\| \overline{\boldsymbol{F}}^{(n)}_{x_{kl} x_{kl}} - \overline{c}_k^{(n)} \right\|_2^2 \tag{25}$$

where $\boldsymbol{x}_{kl}$ represents the lth HRRP in the kth HRRP frame, and L denotes the number of HRRPs in the kth HRRP frame. If the Euclidean distance calculation formula in original HRRP space (3) is used, the HRRP can also be used in the RBFN.

V. Experimental Results

A. Recognition Results for Measured Data

The results presented here are based on measured airplane data. The parameters of targets and radar are shown in Table III, and the projections of target trajectories onto the ground plane are shown in Fig. 4, from which the aspect angle of the airplane can be estimated according to its relative position to radar. As shown in Fig. 4, the measured data are segmented. Training data

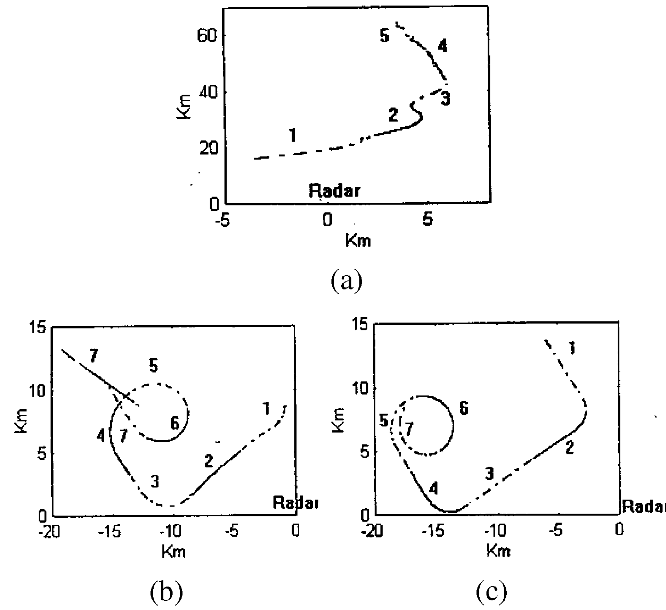

Fig. 4. Projections of target trajectories onto the ground plane. (a) Yark-42. (b) An-26. (c) Cessna Citation S/II.

TABLE III
PARAMETERS OF PLANES AND RADAR IN THE ISAR EXPERIMENT

radar parameters	center frequency		5520 MHz
	bandwidth		400 MHz
planes	length (m)	width (m)	height (m)
Yark-42	36.38	34.88	9.83
An-26	23.80	29.20	9.83
Cessna Citation S/II	14.40	15.90	

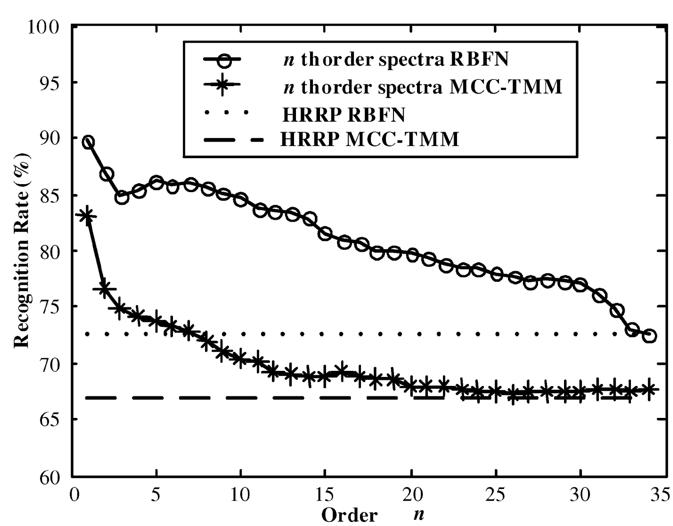

Fig. 5. Recognition rates of higher order spectra features and original HRRP: "$*$" represents the recognition rate of the nth-order spectra feature by MCC-TMM; dashed line represents the recognition rate of the original HRRP by MCC-TMM; "o" represents the recognition rate of the nth-order spectra feature by RBFN; dotted line represents the recognition rate of the original HRRP by RBFN.

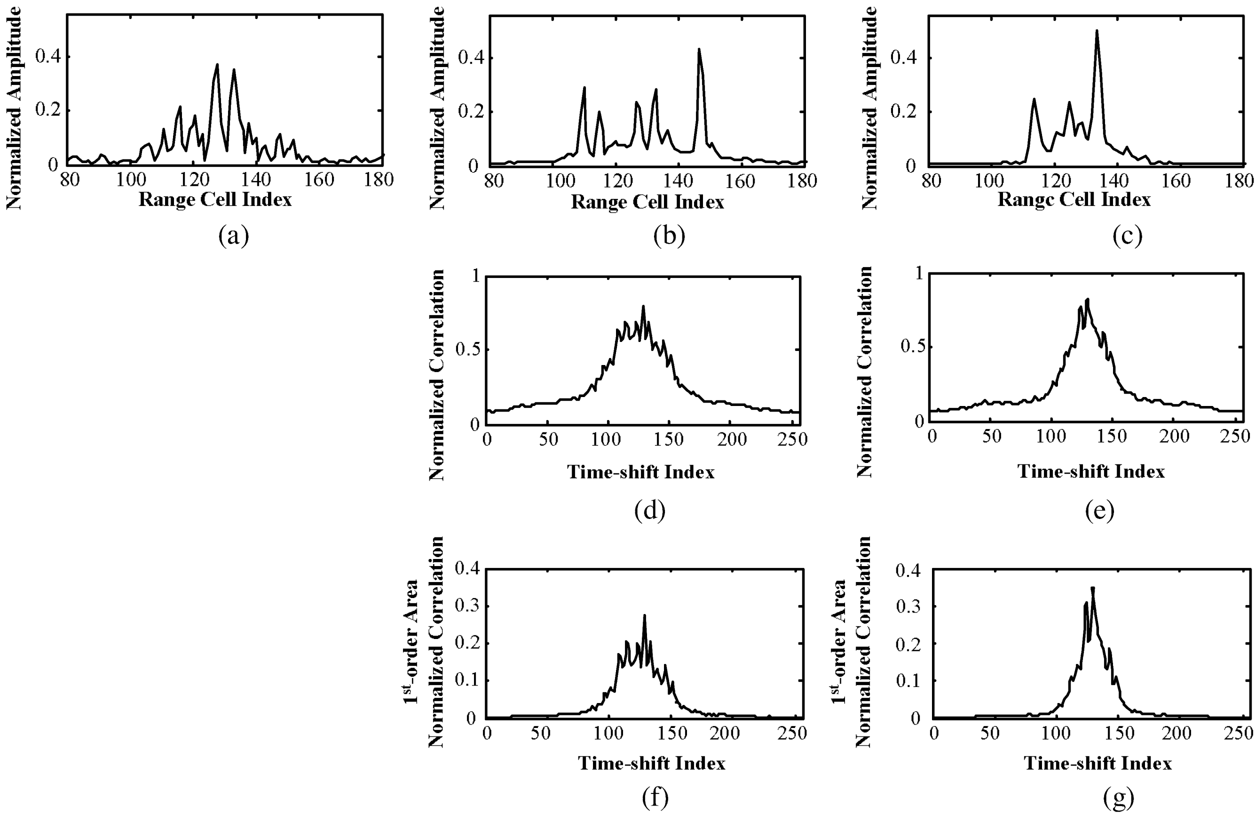

Fig. 6. Example of misclassification in original HRRP space and correct classification in power spectrum feature space. (a) Test data from Yark-42. (b) Template with MCC from Yark-42. (c) Template with MCC from Cessna Citation S/II. (d) Within-class amplitude normalized cross-correlation function. (e) Between-class amplitude normalized cross-correlation function. (f) Within-class 1-order area normalized cross-correlation function. (g) Between-class 1-order area normalized cross-correlation function.

and test data are from different data segments. The second and fifth segments of the Yark-42, the fifth and sixth segments of the An-26, and the sixth and seventh segments of the the Cessna Citation S/II, are taken as the training data, and other data segments are taken as the test data. The training data cover almost all of the target-aspect angles, but their elevation angles are different from those of the test data. According to (14) and the radar parameters shown in Table III, and in view of the few scatterers at the edges of an aircraft-like target and with weak energy, HRRPs in about a 3° angular sector are taken as an HRRP frame in our experiments. Because the measure data we have at hand only include three targets, the experiments performed in the paper are limited to the close-set classification problem, and the false target rejection ability of higher-order spectra features will be evaluated in the future.

In Fig. 5, the recognition rates of higher order spectra features and original HRRP are shown. For both classification algorithms, as the order $n \rightarrow \infty$, the recognition rates of the nth-order spectra feature tend to be the same as those of the original HRRP. The support point can be found from (6). If the order $n \rightarrow \infty$, $\left\langle \boldsymbol{r}_{x_1x_1}^{(n)}, \boldsymbol{r}_{x_2x_2}^{(n)} \right\rangle$ in (6) tends to reach the $(n+1)$th-order power transformed peak value of $r_{x_1x_2}^{(1)}(\tau)$, i.e., $\left\langle \boldsymbol{r}_{x_1x_1}^{(n)}, \boldsymbol{r}_{x_2x_2}^{(n)} \right\rangle \rightarrow \left[\max\limits_{\tau} r_{x_1x_2}^{(1)}(\tau)\right]^{n+1}$. Since power transformation is monotonous, the Euclidean distances in $(n \rightarrow \infty)$th-order spectra feature space and original HRRP space are equivalent when used as pattern similarity measures. Fig. 5 also shows that the recognition rates increase as the order n decreases. Both of the optimal points of the two lines appear at $n = 1$, which corresponds to the power spectrum feature. The recognition rates of the power spectrum feature are $16 \sim 17$ percentage points larger than those of original HRRP. Moreover, the recognition rates by RBFN are larger than those by MCC-TMM by approximately five percentage points. A fact worth emphasizing is that the training and test data are selected from different segments in this paper, but both of them were selected from only one segment of the same measured data in [12]. When the training and test data in [12] are used in our experiment, the recognition rates of bispectra feature by MCC-TMM and RBFN are both larger than 95 percentage points.

B. Analysis of the Classification Performances of Higher Order Spectra Features

The above conclusion can also be analyzed from (4) and (7). The cross-correlation coefficient between two HRRPs is the maximum value of the amplitude normalized cross-correlation function $\overline{r}_{x_1x_2}^{(1)}(\tau)$ [see (4)] and that between two nth-order spectra features is the integral or the area of the area normalized cross-correlation function $\tilde{\gamma}_{x_1x_2}^{(1)}|_{(n+1)}(\tau)$ [see (7)]. Thus, using the cross-correlation coefficient between the nth-order spectra features as the similarity measure between two patterns can avoid misclassification when the high peaks of test data and template are in the same position.

$$\begin{aligned}\left\langle \boldsymbol{r}_{x_1x_1}^{(n)}, \boldsymbol{r}_{x_2x_2}^{(n)} \right\rangle &= \int \cdots \int r_{x_1x_1}(\tau_1,\ldots,\tau_n)\cdot r_{x_2x_2}^{*}(\tau_1,\ldots,\tau_n)d\tau_1\cdots d\tau_n \\ &= \int \cdots \int r_{x_1x_1}(\tau_1,\ldots,\tau_n)\cdot r_{x_2x_2}(\tau_1,\ldots,\tau_n)d\tau_1\cdots d\tau_n \\ &= \int \cdots \int \left[\int x_1(t)x_1(t+\tau_1)\cdots x_1(t+\tau_n)dt\right]\cdot\left[\int x_2(u)x_2(u+\tau_1)\cdots x_2(u+\tau_n)du\right]d\tau_1\cdots d\tau_n \\ &= \int\int x_1(t)x_2(u)\left[\int x_1(t+\tau)x_2(u+\tau)d\tau\right]^n dtdu. \end{aligned} \tag{A.1}$$

An example of misclassification in the original HRRP space and correct classification in the power spectrum feature space is shown in Fig. 6. Because the high peaks in Fig. 6(a) and (c) are almost in the same position, the between-class MCC, as shown in Fig. 6(e), is large than the within-class MCC, as shown in Fig. 6(d), which leads to a misclassification in the original HRRP space. However, the integral of Fig. 6(f) is larger than that of Fig. 6(g); thus, a correct classification is obtained in power spectrum feature space. Although that shown in Fig. 6 is just a particular case, many similar cases occurred in our experiment. This can explain why the recognition performance of the power spectra is better than that of the original HRRP to some extent.

VI. Conclusion

A method for calculating the Euclidean distance in higher order spectra feature space is proposed in this paper. This method is performed in the original HRRP space directly, which avoids calculating the higher order spectra; therefore, the computation complexity and storage requirement can be extremely decreased. Based on the scattering center model, a robust template database establishment method is presented, i.e., using the feature vectors extracted from average profiles as feature templates in HRRP-based RATR. Physical mechanism of this method is analyzed in detail in this paper. The recognition results for measured data show that the power spectrum feature has a better recognition performance than the other higher order spectra features.

Appendix
Inner Product of Higher Order Autocorrelation Functions

Let $\boldsymbol{r}_{x_1x_1}^{(n)}$ and $\boldsymbol{r}_{x_2x_2}^{(n)}$ represent the higher order autocorrelation functions of two real and continuous functions $x_1(t)$ and $x_2(t)$, respectively. Find out their inner product in (A.1), shown at the top of the page. Do the variable substitution. Let $t+\tau=v$, $u-t=s$, $t=t'$. Rewrite the above equation as

$$\begin{aligned}&\left\langle \boldsymbol{r}_{x_1x_1}^{(n)}, \boldsymbol{r}_{x_2x_2}^{(n)} \right\rangle \\ &\quad=\int\left[\int x_1(t')x_2(t'+s)dt'\right]\left[\int x_1(v)x_2(v+s)dv\right]^n ds \\ &\quad=\int\left[\int x_1(v)x_2(v+s)dv\right]^{n+1} ds \\ &\quad=\int\left[r_{x_1x_2}^{(1)}(\tau)\right]^{n+1} d\tau. \end{aligned} \tag{A.2}$$

For discrete signals, the corresponding equation is

$$\left\langle \boldsymbol{r}_{x_1x_1}^{(n)}, \boldsymbol{r}_{x_2x_2}^{(n)} \right\rangle = \sum_m \left[r_{x_1x_2}^{(1)}(m)\right]^{n+1}. \tag{A.3}$$

Acknowledgment

The authors wish to thank Associate Editor Prof. A. Gershman and the three anonymous reviewers for their valuable comments and suggestions, which have helped to significantly improve this manuscript. In addition, L. Du wishes to thank Prof. J. Zhang and Prof. Q. Du for their valuable comments and suggestions.

References

[1] R. A. Mitchell and R. Dewall, "Overview of high range resolution radar target identification," in *Proc. Autom. Target Recognition Working Group*, Monterey, CA, 1994.
[2] H.-J. Li and S.-H. Yang, "Using range profiles as feature vectors to identify aerospace objects," *IEEE Trans. Antennas Propag.*, vol. 41, no. 3, pp. 261–268, Mar. 1993.
[3] A. Zyweck and R. E. Bogner, "Radar target classification of commercial aircraft," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 32, no. 2, pp. 598–606, 1996.
[4] R. Williams, J. Westerkamp, D. Gross, and A. Palomino, "Automatic target recognition of time critical moving targets 1D high range resolution (HRR) radar," *IEEE AES Syst. Mag.*, vol. 15, no. 4, pp. 37–43, 2000.
[5] R. A. Mitchell, "Robust High Range Resolution Radar Target Identification using a Statistic Feature Based Classifier with Feature Level Fusion," Ph.D. dissertation, Univ. Dayton, Dayton, OH, 1997.
[6] S. P. Jacobs, "Automatic Target Recognition using High-Resolution Radar Range Profiles," Ph.D. dissertation, Washington Univ., St. Louis, MO, 1999.
[7] R. Van der Heiden and F. C. A. Groen, "The box-cox metric for nearest neighbor classification improvement," *Pattern Recogn.*, vol. 30, no. 2, pp. 273–279, 1997.
[8] X. Liao, "Radar Target Recognition using High Resolution Range Profiles," Ph.D. dissertation, Xidian Univ., Xi'an, China, 1999.
[9] X. Liao, Z. Bao, and M.-D. Xing, "On the aspect sensitivity of high resolution range profiles and its reduction methods," in *Conf. Rec. IEEE Int. Radar*, May 2000, pp. 310–315.
[10] M.-D. Xing, Z. Bao, and B. Pei, "The properties of high-resolution range profiles," *Opt. Eng.*, vol. 41, no. 2, pp. 493–504, 2002.
[11] X. Liao and Z. Bao, "Circularly integrated bispectra: Novel shift invariant features for high-resolution radar target recognition," *Electron. Lett.*, vol. 34, no. 19, pp. 1879–1880, 1998.
[12] X. Zhang, Y. Shi, and Z. Bao, "A new feature vector using selected bispectra for signal classification with application in radar target recognition," *IEEE Trans. Signal Process.*, vol. 49, no. 9, pp. 1875–1885, Sep. 2001.
[13] W. G. Carrara, R. S. Goodman, and R. M. Majewski, *Spotlight Synthetic Aperture Radar —Signal Processing Algorithms*. Norwood, MA: Arthech House, 1995.
[14] V. Chandran and S. L. Elgar, "Pattern recognition using invariants defined from higher order spectra—One-dimensional inputs," *IEEE. Trans. Signal Process.*, vol. 41, no. 1, pp. 205–212, Jan. 1993.

[15] J. K. Tugnait, "Detection of non-Gaussian signals using integrated polyspectrum," *IEEE Trans. Signal Process.*, vol. 42, no. 11, pp. 3137–3149, Nov. 1994.
[16] C. L. Nikias and A. P. Petropulu, *High-Order Spectra Analysis: A Nonlinear Signal Processing Framework*. Englewood Cliffs, NJ: Prentice-Hall, 1993.
[17] J. M. Mendel, "Tutorial on higher order statistics (spectra) in signal processing and system theory: Theoretical results and some applications'," *Proc. IEEE*, vol. 79, no. 3, pp. 278–305, Mar. 1991.
[18] V. Popovici and J. P. Thiran, "Pattern recognition using higher-order local autocorrelation coefficients," in *Proc. IEEE Worshop Neural Networks Signal Process.*, vol. 12th, 2002, pp. 229–238.
[19] V. N. Vapnik, *The Nature of Statistical Learning Theory*. New York: Springer-Verlag, 1995.
[20] Z. Bao, C.-Y. Sun, and M.-D. Xing, "Time-frequency approaches to ISAR imaging of maneuvering targets and their limitations," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 3, pp. 1091–1099, Mar. 2001.
[21] S. A. Gorshkov, S. P. Leschenko, V. M. Orlenko, S. Y. Sedyshev, and Y. D. Shirman, *Radar Target Backscattering Simulation Software and User's Manual*. Boston, MA: Artech House, 2002.
[22] Y. D. Shirman, *Computer Simulation of Aerial Target Radar Scattering, Recognition, Detection, and Tracking*. Boston, MA: Artech House, 2002.
[23] T. G. Dietterich, "Approximate statistical tests for comparing supervised classification learning algorithms," *Neural Comput.*, vol. 10, pp. 1895–1923, 1998.
[24] J. P. Marques, *Pattern Recognition Concepts, Methods and Applications*. New York: Springer, 2001.
[25] K. Fukunaga, *Introduction to Statistical Pattern Recognition*, Second ed. Boston, MA: Academic, 1990.
[26] J. T. Tou and R. C. Gonzalez, *Pattern Recognition Principles*. London, U.K.: Addison-Wesley, 1974.

Lan Du received the B.Eng. and M.Eng. degrees in electronic engineering from Xidian University, Xi'an, China, in 2001 and 2004, respectively. She is currently working toward the Ph.D. degree with the National Lab of Radar Signal Processing, Xidian University.

Her research interests are in the fields of radar automatic target recognition (RATR), radar imaging (SAR/ISAR), and radar signal processing.

Hongwei Liu (M'00) received the M.S. and Ph.D. degrees in electronic engineering from Xidian University, Xi'an, China, in 1995 and 1999, respectively.

He was with the National Lab of Radar Signal Processing, Xidian University. From 2001 to 2002, he was a visiting scholar with the Department of Electrical and Computer Engineering, Duke University, Durham, NC. He is currently an associated professor at National Lab. of Radar Signal Processing, Xidian University. His research interests are radar automatic target recognition, radar signal processing, and adaptive signal processing.

Zheng Bao (M'80–SM'90) graduated from the Communication Engineering Institution of China, Xi'an, in 1953.

Currently, he is a professor with Xidian University, Xi'an, and a member of the Chinese Academy of Science. He is the author or co-author of six books and has published more than 300 papers. His research work focuses on the areas of space-time adaptive processing (STAP), radar imaging (SAR/ISAR), radar automatic target recognition (RATR), over-the-horizon radar (OTHR) signal processing, and passive coherent location (PCL).

Mengdao Xing was born in Zhejiang, China, in 1975. He received the B.S. and Ph.D. degrees in 1997 and 2002, respectively, from Xidian University, Xi'an, China.

He is now a professor with the National Lab of Radar Signal Processing, Xidian University. His research interests are radar imaging, target recognition, and over the horizon radar (OTHR) signal processing.